Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - VII

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PART – A Theory

1. Import Matplotlib

```
import matplotlib.pyplot as plt
```

`pyplot` is used to create and display different types of charts.

2. Scatter Plot

- Used to show the relationship between two numerical values.
- Function: `plt.scatter(x, y)`
- Example: Study hours vs. Marks.

3. Bar Chart

- Used to compare values between different categories.
- Function: `plt.bar(x, y)`
- Example: Marks of different students.

4. Histogram

- Used to show the distribution of numerical data.
- Function: `plt.hist(data)`
- Example: Distribution of students' marks.

5. Pie Chart

- Used to show parts of a whole.
- Function: `plt.pie(values, labels=labels)`
- Example: Monthly expenses.

6. Displaying the Chart

```
plt.show()
```

`plt.show()` displays the created chart.

Common Functions

```
plt.title("Chart Title")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

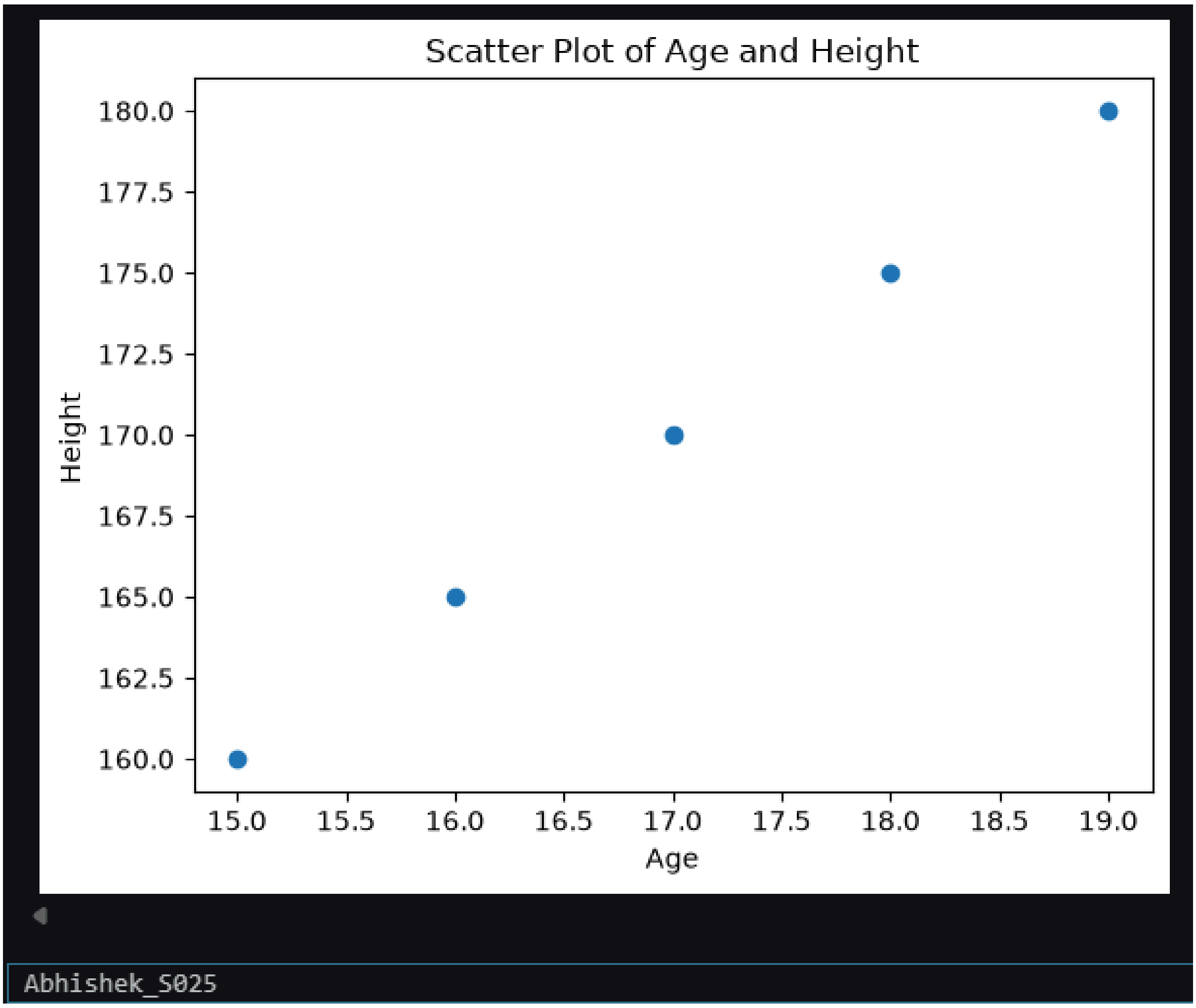
PART – B Practical Questions

a. Write a Python program to create a scatter plot of Age and Height of 5 students.

CODE :

```
import matplotlib.pyplot as plt
import numpy as np

age = [15, 16, 17, 18, 19]
height = [160, 165, 170, 175, 180]

plt.scatter(age, height)

plt.xlabel("Age")
plt.ylabel("Height ")
plt.title("Scatter Plot of Age and Height")

plt.show()
print("Abhishek_S025")
```

OUTPUT :

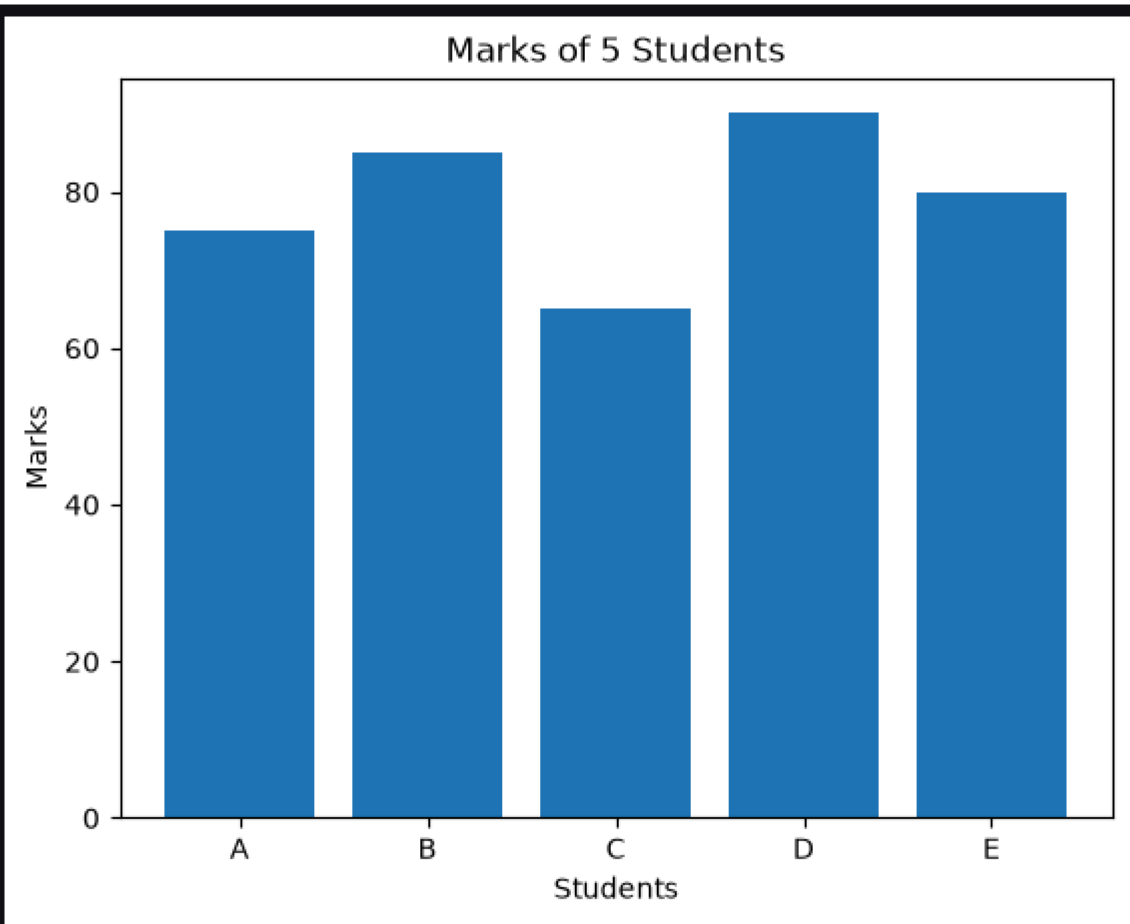


b. Write a Python program to create a bar chart showing marks of 5 students.

CODE :

```python
import numpy as np

students = ["A", "B", "C", "D", "E"]
marks = [75, 85, 65, 90, 80]

plt.bar(students, marks)

plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Marks of 5 Students")

plt.show()
print("Abhishek_S025")
```

OUTPUT :



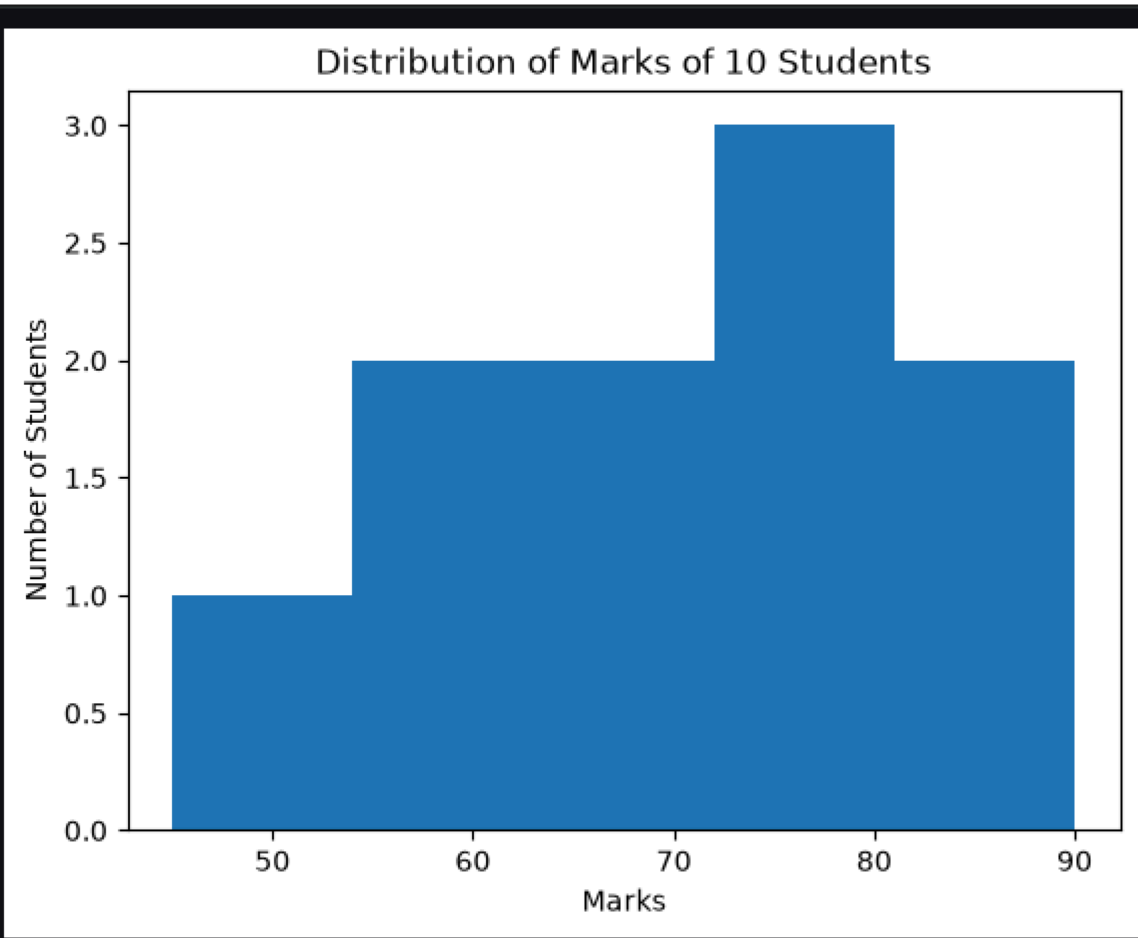

```
Abhishek_S025
```

c. Write a Python program to create a histogram showing the distribution of marks of 10 students.

CODE :

```python
import numpy as np

marks = [45, 55, 60, 65, 70, 72, 75, 80, 85, 90]

plt.hist(marks, bins=5)

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Marks of 10 Students")

plt.show()
print("Abhishek_S025")
```

OUTPUT :



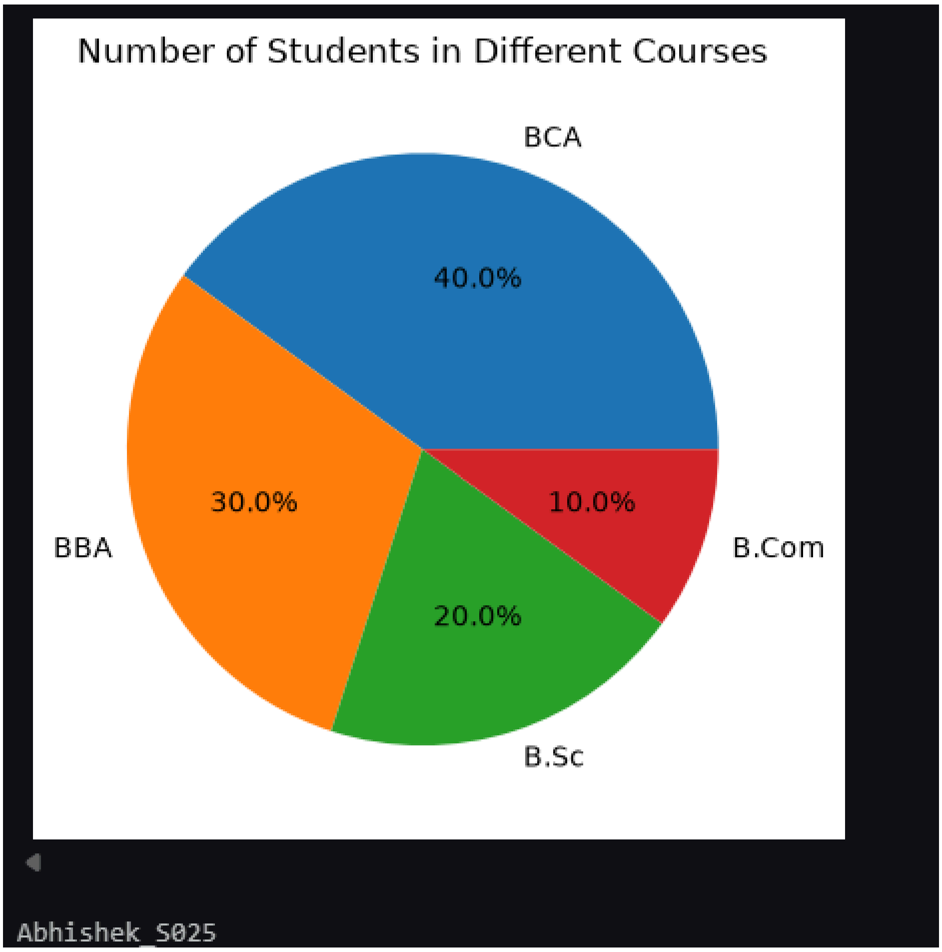

```
Abhishek_S025
```

d. Write a Python program to create a pie chart showing the number of students in different courses.

CODE :

```
import numpy as np

courses = ["BCA", "BBA", "B.Sc", "B.Com"]
students = [40, 30, 20, 10]

plt.pie(students, labels=courses, autopct="%1.1f%%")

plt.title("Number of Students in Different Courses")

plt.show()
print("Abhishek_S025")
```

OUTPUT :

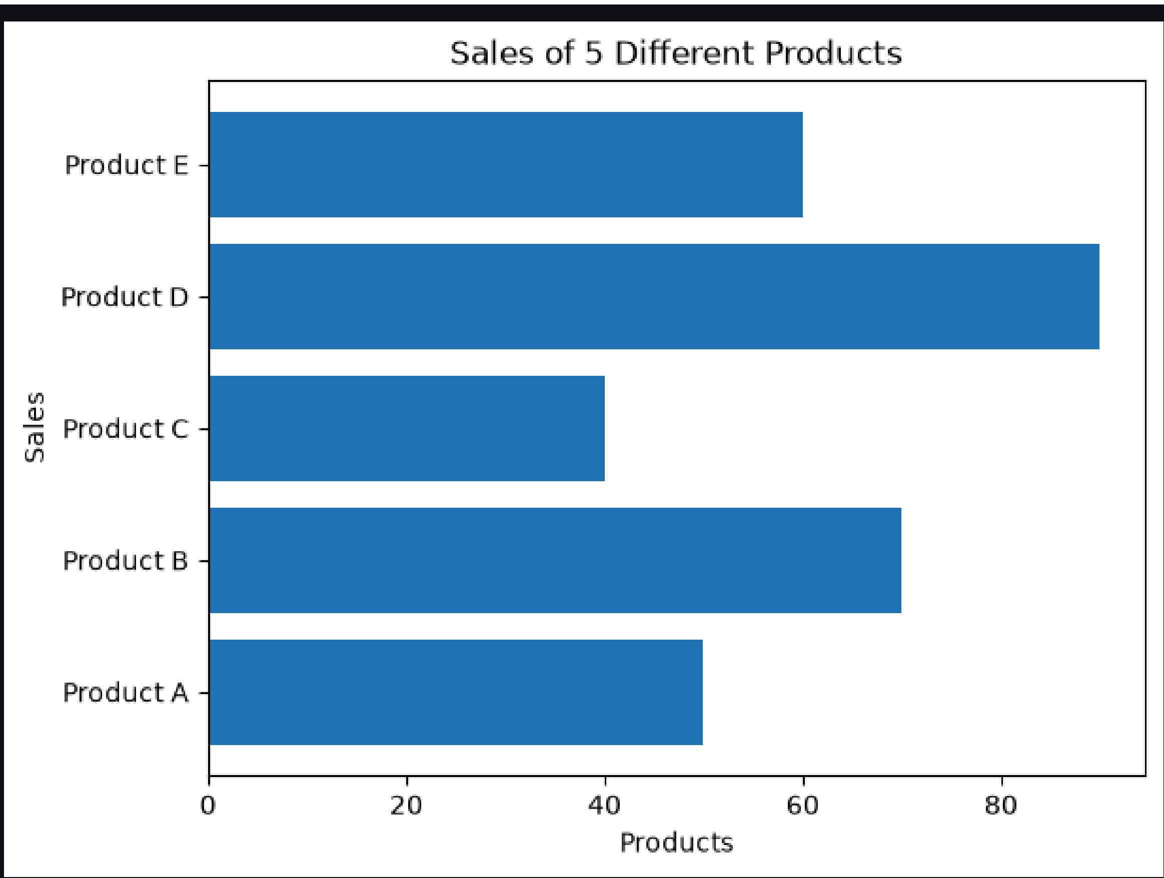


e. Write a Python program to create a bar chart showing sales of 5 different products.

CODE :

```
import numpy as np

products = ["Product A", "Product B", "Product C", "Product D", "Product E"]
sales = [50, 70, 40, 90, 60]

plt.barh(products, sales)

plt.xlabel("Products")
plt.ylabel("Sales")
plt.title("Sales of 5 Different Products")

plt.show()
print("Abhishek_S025")
```

OUTPUT :




```
Abhishek_S025
```